



— DEFENSE & NATIONAL SECURITY

IGNITING DEFENSE INNOVATION THROUGH SCIENCE-POWERED SOLUTIONS.

SCIENCE AND TECHNOLOGY WITH PASSION

We turn science into mission advantage for those who defend. SSAI delivers mission-critical science, engineering, and technology to the warfighters, operators, and analysts who protect the nation. The same engineering trusted on NASA's flagship missions now brings zero-trust cybersecurity, AI/ML, a cleared Mil-spec manufacturing base, deep-space operations, and instrument engineering to the defense enterprise, engineered to work the first time, and every time.

1977

FOUNDED · 49+ YEARS

350+

SCIENTISTS &
ENGINEERS

TS FCL

TOP SECRET FACILITY
CLEARANCE

42K

SQ FT CLEARED
MANUFACTURING &
MACHINE SHOP

100+

MAJOR INSTRUMENTS,
MODELS & MISSIONS

3

WHOLLY-OWNED
SUBSIDIARIES

DEFENSE CAPABILITIES

ENGINEERED FOR THE MISSIONS THAT CANNOT FAIL.

We serve the warfighters, operators, and analysts who defend the nation, with a cleared, vertically integrated base spanning digital defense, cleared manufacturing, space operations, and mission-grade engineering.

- 01**
CYBERSECURITY & DIGITAL DEFENSE
We make platforms smarter and secure, so operators stay ahead of the threat. Advanced IT and cybersecurity on zero-trust architectures, with AI/ML for anomaly detection, autonomy, and cross-agency support. Our CREST framework (Coupled Reusable Earth System Tensor) is an AI-enabled Python framework for operational digital twins, taking the mission from sensor to decision.
- 02**
SATELLITE & SPACE SYSTEMS MANUFACTURING
We build the cleared, flight-grade hardware the mission depends on. A 32,000 sq ft cleared electronics manufacturing facility (PCB design, fab, and test; FPGA board repair; ESD-protected floor; three clean rooms; cleared personnel) and a 10,000 sq ft Mil-spec machine shop (Mil-I-45208A, Mil-std-45662; CNC, lathes, presses, mills; certified aerospace machinists) under two miles from NASA Goddard.
- 03**
SPACE OPERATIONS
We keep operators ahead in a contested domain. 24/7 operational modeling, deep-space instrument development and calibration, trajectory design and atmospheric-entry modeling, and mission operations from pre-launch through on-orbit, space expertise that transfers directly to defense and resilient space.
- 04**
MISSION-GRADE ENGINEERING
We engineer the hardware that has to work the first time, and every time. Systems engineering, requirements and V&V, avionics and FPGA design, software engineering (CMMI L3), EEE parts processing, and system integration and testing, to AS9100, ITAR registration, and J-STD-001F workmanship, the rigor that flies NASA flagships, applied to defense.
- 05**
ENVIRONMENTAL INTELLIGENCE
We turn what sensors see into decision advantage for the operator. Data assimilation, remote sensing, weather and space-weather forecasting, and multi-source fusion for ISR and operational decisions when the mission cannot wait.

MISSION ALIGNMENT

ALIGNED TO THE DEFENSE PORTFOLIOS THE MISSION NEEDS.

AI / ML CREST digital twins, anomaly detection, custom AI.	AUTONOMY Autonomous sensing, modeling, and decision support.	CYBER & TELECOM Zero-trust architectures, secure IT and comms.	SPACE Space operations, instruments, resilient space.	EMERGING TECHNOLOGY FPGA, digital twin, advanced analytics.	ENERGY Environmental modeling and resource intelligence.	HUMAN SYSTEMS Decision-grade data products and visualization.
--	--	--	---	---	--	---

Top Secret Facility Clearance

UEI PJ2SP76D3HJ3

CAGE 5S009

NAICS 541715 (small) / 541330

ITAR Registered · WOSB

AS9100 · CMMI L3 · ISO 9001

DIFFERENTIATORS

A PARTNER THE MISSION CAN TRUST.

- **Cleared for the classified mission.** We hold a Top Secret Facility Clearance with cleared facility and cleared personnel, ITAR registered, ready for classified work.
- **A cleared, Mil-spec manufacturing base.** We run 42,000 sq ft of in-house electronics manufacturing and precision machining under two miles from NASA Goddard.
- **In the D.C. defense corridor.** We work minutes from NASA Goddard and the Washington defense community, where requirements are written and missions are run.
- **Aerospace-grade engineering behind every delivery.** AS9100, CMMI L3 for Development, ISO 9001:2015, EEE parts processing, and J-STD-001F workmanship.

Top Secret FCL

ITAR Registered

AS9100 Aerospace Quality

CMMI L3 Development

ISO 9001:2015

SUBSIDIARIES

ONE TRUSTED SOURCE FOR THE MISSION.

Three wholly-owned subsidiaries put the capability the defense mission needs under one roof.

ADVANCED MISSION PARTNERS
Electronics manufacturing, engineering services, EEE expertise, and precision manufacturing.

ELUCIDATION CONCEPTS, LLC
Cybersecurity solutions, systems engineering, system integration testing, and test & evaluation.

SIGMA ENGINEERING & FABRICATION, INC.
Precision machining, manufacturing, material expertise, and quality at speed.

PAST PERFORMANCE

TRUSTED ON THE MISSIONS THAT CANNOT FAIL.

ESES III	NASA Goddard, Engineering Technology Directorate (CPAF). Study, design, development, fabrication, integration, testing, verification, and operations of spaceflight, airborne, and ground hardware and software. 11/2018 to 04/2025.
HBG	Hydrospheres, Biospheres and Geophysics. NASA Goddard, Earth Sciences Division (CPFF). Scientific research across space, airborne, in-situ, and laboratory observations. 04/2020 to 11/2025.
STARSS III	Science, Technology and Research Support Services. NASA Langley, Science Directorate (CPFF). Earth atmosphere and climate, extending to other planetary atmospheres. 12/2016 to 05/2023.
SAMDA	Support for Atmospheric Modeling and Data Assimilation. NASA Goddard, Global Modeling and Assimilation Office (CPFF). 03/2017 to 11/2025.
HBS	Hydrospheric and Biospheric Sciences. NASA Goddard, Earth Sciences Division (CPFF). 12/2014 to 05/2020.

Flagship missions: OCI / PACE · Roman Space Telescope · Lucy · Mars Sample Return / CCRS · Dragonfly · GeoXO · DAVINCI+ · TSIS-2 · JWST.

CORPORATE DATA

THE ESSENTIALS.

UEI	PJ2SP76D3HJ3	CONFIRM	CAGE	5S009	CONFIRM
NAICS	541715 (small) / 541330		BUSINESS STATUS	WOSB, SBA 2024	CONFIRM
GSA MAS	47QRAA23D0014	CONFIRM	IDIQ VEHICLE	OASIS Small Business	CONFIRM
SECURITY	Top Secret FCL, ITAR Registered		QUALITY & MATURITY	AS9100, CMMI L3, ISO 9001:2015	
FOUNDED	1977		HEADQUARTERS	Lanham, MD	

BUSINESS DEVELOPMENT

BD@ssaihq.com
(301) 867-2000

HEADQUARTERS

The Aerospace Building
10210 Greenbelt Rd, Suite 600
Lanham, MD 20706

LOCATION ADVANTAGE

Minutes from NASA Goddard, in the Washington D.C. defense corridor.

Procurement identifiers and contract values marked "confirm" / "SSAI to confirm" are pending SSAI sign-off before external release.